VIJAYALAKSHMI S

Biomedical Engineer | Research & Clinical Technology Specialist

Chennai, India • +91 9884305439 • vijisb24@gmail.com • linkedin.com/in/vijaylakshmi-sivaraman

PROFESSIONAL SUMMARY

Results-driven Biomedical Engineer with international research experience spanning Germany and India, specialising in **wearable medical device development**, **advanced MRI image processing**, and **clinical data analysis**. At the German Cancer Research Center (DKFZ), developed a MATLAB-based pipeline for CEST MRI at 9.4 T that improved SNR by **500%** and cut post-processing time by **40%**. Currently leading an ANRF-funded **IMU-based gait monitoring system** at Anna University — work recognised with the **Best Paper Award at MedNext 2025**. Ranked **AIR 270 in GATE 2026** (Biomedical Engineering), reflecting strong foundations in biomedical systems and instrumentation. Eager to contribute technical rigour and cross-functional collaboration skills to a forward-thinking organisation in MedTech or biomedical research.

CORE COMPETENCIES

Wearable Sensor Systems • IMU & Signal Processing • Medical Device Validation • CEST MRI Image Analysis • MATLAB & Python Development • Real-Time Data Visualisation • Embedded Systems • Machine Learning (Scikit-Learn) • Regulatory Documentation (DST-ANRF) • Project Planning & Milestone Tracking • Stakeholder Communication • Technical Documentation

RESEARCH & WORK EXPERIENCE

Project Associate — ANRF Funded Project *Jan 2025 – Present*

Centre for Medical Electronics, Anna University | Chennai, India

- Led end-to-end design of an **IMU-based gait monitoring system**, achieving **98% signal precision** in real-time gait analysis.
- Developed a **Python application** for real-time gait visualisation enabling direct clinical validation with physiotherapists and MedTech partners.
- Managed procurement, budgeting, and **DST-ANRF regulatory compliance** documentation, ensuring on-schedule project milestones.
- Presented prototypes at **Medicall 2025**; engaged clinicians and MedTech industry stakeholders to validate clinical utility.

Research Assistant *Nov 2022 – Sep 2023*

German Cancer Research Center (DKFZ) | Heidelberg, Germany

- Built a **MATLAB pipeline for CEST MRI** at 9.4 T, improving **SNR by 500%** using Lorentzian six-pool spectral fitting.
- Streamlined MRI post-processing workflows with physicists and data scientists, **reducing processing time by 40%**.
- Conducted preclinical mouse brain imaging studies contributing to tumour microenvironment characterisation research.

KEY PROJECTS

Low-Cost Wearable Ankle Angle Measurement System ◆ Best Paper — MedNext 2025

Python · Embedded Systems · IMU Sensors

- Engineered an IMU wearable for ankle-angle tracking with real-time Python visualisation; optimised cost below **■2,000**.
- Recognised with **Best Paper Award at MedNext 2025** for innovation in low-cost accessible biomechanics.

CEST MRI Post-Processing Optimisation (M.Sc. Thesis / DKFZ)

MATLAB · Image Processing · MRI Analysis

- Proposed a two-step workflow eliminating magnetisation transfer spillover, **improving reproducibility by 20%**.

Healthcare ML Classifier — IBM Data Science Capstone

Python · Scikit-Learn · EDA · REST APIs

- Built and tuned a Decision Tree classifier on healthcare data achieving **83% accuracy**; visualised insights with Matplotlib.

EDUCATION

M.Sc. Biomedical Engineering *2021 – 2023*

Heidelberg University | Germany

Thesis: Raw-data reconstruction workflow for preclinical CEST MRI at 9.4 T

B.E. Biomedical Engineering *2017 – 2021*

Rajalakshmi Engineering College, Anna University | Chennai, India

Thesis: 3D skin modelling and SAR analysis using COMSOL Multiphysics

TECHNICAL SKILLS

Programming & Analysis: Python, MATLAB, R, SQL

Data Science & ML: NumPy, Pandas, Scikit-Learn, Matplotlib, EDA, Decision Trees

Biomedical Systems: Medical device validation, IMU sensors, signal processing, wearable devices, embedded systems

Imaging & Simulation: CEST MRI, MRI image processing, Lorentzian fitting, COMSOL Multiphysics

Project Management: Project planning support, milestone tracking, schedule coordination, task prioritisation

Professional: Technical documentation, regulatory compliance (ANRF/DST), stakeholder communication

AWARDS & CERTIFICATIONS

GATE 2026 — Biomedical Engineering (BM) — All India Rank (AIR) **270** — Graduate Aptitude Test in Engineering, 2026

Best Paper Award — MedNext 2025 — Low-Cost Wearable Ankle Angle Measurement Device — innovation in accessible biomechanics

IBM Data Science Professional Certificate — Python, Data Analytics, Visualisation, Machine Learning — IBM / Coursera, 2024

Data Analytics with R Programming — Statistical Data Analysis in R, 2024